

Session-09: Exponential Distribution

1. If X is an exponential variate with mean 5 find

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| (i) | $P(0 < x < 1)$ | Ans: 0.1813 |
| (ii) | $P(-\infty < x < 10)$ | Ans: 0.8646 |
| (iii) | $P(x \leq 0 \text{ or } x \geq 1)$ | Ans: 0.8187 |

2. The time in minutes X between the arrival of successive customers at a post office is exponentially distributed with pdf $f(x) = 0.2 e^{-0.2x}$

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| i. | What is the expected time between arrivals? | Ans: 5 minutes |
| ii. | A customer walks into the post office at 12.30 pm. What is the probability the next customer arrives | |
| | (a) on or before 12.32 pm.? | Ans: 0.330 |
| | (b) after 12.35 pm.? | Ans: 0.368 |

3. The average rate of job submissions in a busy computer Centre is 4 per minute. If it can be assumed that the number of submissions per minute interval is Poisson distributed, calculate the probability that:

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| (a) | at least 15 seconds will elapse between any two jobs. | Ans: 0.3679 |
| (b) | less than 1 minutes will elapse between jobs. | Ans: 0.9817 |